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Policies enhancing agricultural value chains in developing countries: An Evidence Gap Map

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ABSTRACT

Strengthening agricultural value chains (AVCs) and ensuring continuous supply of food is of upmost importance especially for developing countries where food security and malnutrition remain vital threats. However, effective and efficient coordination of AVCs in developing countries continues to remain an issue due to capacity constraints and inefficiencies of interventions. An Evidence Gap Map (EGM) is used to systematically retrieve, organize, and map existing knowledge on AVCs and identify gaps in the literature that will suggest future areas for effective policy interventions. The marginal increase in knowledge of a project focusing on an area of a knowledge gap may be larger than that of a project which focuses on an area with many studies. A total of 4745 studies were retrieved from a systematic search, however only 165 studies meeting inclusion criteria are included in EGM. Most policy interventions address inputs supply stage of value chains with specific focus on subsidies and agricultural loans. Governance and issues related to gender, poverty and social groups are underexplored. Most frequently mentioned outcomes subcategories were yield, agricultural profits, family income, and technology adoption. Outcomes related to risk management and product movement in time and space had the smallest frequency of occurrence. As noted, underexplored areas provide opportunities for future research.

1. Introduction

Ensuring availability of and accessibility to food is crucial in developing (low, middle, and upper-middle income) countries where food security and malnutrition are critical concerns [1,2]. Improving the agriculture sector, which also plays an important role in driving economic growth [3], will contribute to increasing food security and decreasing malnutrition [2,4]. Further, promoting agricultural growth in developing countries is important not only for their own development, but also for the well-being of the world, as these countries produce nearly half of the global food supply [5]. Improvements in the agricultural value chains (AVCs) hold promise in promoting growth and development and helping alleviate food insecurity [2].

Value chains are

"… the full range of activities which are required to bring a product or service from conception, through the different phases of production (involving a combination of physical transformation and the input of various producer services), delivery to final consumers, and final disposal after use" [6].

In AVCs, activities occur in a sequence of stages, involving transformation of the agricultural product in time, space, or physical form. Effective and efficient coordination of AVCs, however, continues to remain an issue in developing countries [7–9]. Given the importance of agriculture, it is not surprising that numerous studies have been conducted on AVCs which have led to a considerable number of reviews. These reviews, most of which are systematic in nature [10–15], provide valuable information by gathering and summarizing existing evidence on a specific question or value chain activity. Although in-depth, this information is generally narrow in focus [16]. In addition, systematic reviews often fail to provide policy-relevant advice because of a lack of high-quality primary evidence [17]. Following along the lines of Sutton et al. [18], who note the importance of determining where we have been

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and where we may go, this study takes a broader approach to understanding AVCs. As opposed to systematic reviews, this approach does not summarize what the evidence says, but rather shows what evidence is available.

The specific objective of this study is to identify existing evidence on interventions in AVCs and to emphasize gaps in our knowledge in the AVCs. This objective is achieved by mapping existing studies through systematic screening of the literature on AVCs. The resulting map provides a visual representation of the frequencies of studies that have been conducted on a specific topic; therefore, allowing organizations and agencies to target knowledge gaps potentially leading to increased knowledge concerning the policy interventions.

Evidence Gap Maps (EGMs), a complementary method to systematic reviews, is used to identify gaps in research, using both systematic reviews and primary studies. Further, by identifying gaps in reviews, EGMs can also help identify where systematic reviews are necessary and could "… add particular value" [17]. Being part of the so called "big picture" family of reviews, EGMs have a broader scope as compared to systematic reviews [19]. Being "… a relatively new approach to systematically identify and report the range of research activity in broad topic areas or policy domains" [20], EGMs are gaining in popularity as shown by Phillips et al. [21], identifying 73 maps for international development. One reason for their popularity is EGMs provide guidance for researchers and funding agencies as to where gaps in the knowledge base exist allowing potentially for a more efficient focus of their efforts. Essentially, EGMs serve as a tool to support informed policy making [22], as they summarize what evidence is available on a certain topic.

This study has several contributions to AVC-related policymaking. First, the EGM informs decision makers on the availability of primary research of agricultural policies across stages of value chains and reveals gaps in knowledge to guide future research and programs. Second, based on the results of the EGM, policymakers may want to reallocate funding to underexplored areas. Third, researchers can use the information to identify areas where further research is necessary to advance knowledge in those domains.

2. Conceptual framework

The conceptual model guiding the mapping of studies to identify knowledge gaps is displayed in Fig. 1. This framework links policy interventions with outcomes and longer-term impacts through the AVC. The aims of interventions in the AVC are to generate and sustain longer term improvements which are categorized into performance, competitiveness, resilience, and sustainability. Competitiveness interlinks quantity, quality, and reliability in supply to reducing cost per unit of product [23]. Performance refers to economic and financial related outcomes [15]. Resilience is the ability to quickly recover from disruption [24]. Sustainability integrates economic and managerial, social, environmental, and political components characterizing the value chain [25].

At the top of the conceptual model are policy interventions which are defined as any activity implemented by national or international authorities and non-state actors [26]. Interventions involve regulations, infrastructure provision or improvements, market incentives and information that address a variety of issues including technology, processes, behavior, and other practices of AVC stakeholders. The AVC is divided into five stages, input, production/harvest, transport and storage, processing, and marketing. Interventions targeting these stages are linked to outputs through actions of stakeholders in the AVC. Although interventions may address multiple stages, they are classified according to the stage that is their primary focus. Interventions associated with overall governance and inclusion are considered a separate stage of the AVC. Intervention categories and examples by AVC stage are provided in Table 1.

Changes in outcomes ultimately influence the ability of the agricultural sector to achieve its long-term goals. Since outcomes are diverse and not necessarily associated with an AVC stage, they are grouped into five categories: production, market, risk management, product movement, and welfare. Each category of outcomes is composed of subcategories which are presented in Table 2.

Using the conceptual model, a matrix is developed for the EGM in which the rows are interventions (by subcategory) and columns are outcomes (by subcategory) with the resulting cells indicating the frequency of studies for that intervention/outcome combination. The EGM does not indicate the study results concerning the relationship between the intervention and an outcome. Instead, it identifies the number of studies on each intervention/outcome pair, thus highlighting gaps in research.

3. Methods

3.1. Search strategy

A database search protocol to determine the inclusion and exclusion criteria for the EGM was developed following the Population, Interventions, Comparison, Outcomes, and Study Design (PICOS) framework [27]. An academic librarian specialized in search syntheses adopted the protocol and conducted the search in April 2022 (see Appendix A for search keywords). The search was implemented through major databases focused on agricultural policy and research: Agricola, Business Source Ultimate, Academic Search Ultimate, Cab Abstracts, and EconLit with full text. These databases provide the most efficient coverage of the AVC literature range of economics, business, and agriculture, without returning many irrelevant references.

In addition, manual searches of grey literature including fifteen development organization websites[1] were conducted. As a result, a wide range of studies, including journal articles, working papers, conference presentations, theses/dissertations, book chapters, and reports from research institutions are incorporated. Studies include both primary research and systematic reviews.

Both academic published and grey literature are included to capture as much evidence as possible. Including both type of literature is important since many agricultural policy related research and reports are found on intergovernmental organizations (FAO, World Bank, USAID, etc.) websites and not necessarily designed to be published in journals. Nevertheless, as discussed below, the inclusion criteria required robust methods had to be used regardless of the literature source.

3.2. Study eligibility criteria

The objective is addressed using evidence of interventions in AVCs from primary research and systematic reviews that provide information on impact of interventions (the word "study" refers to both primary articles and reviews). Specific details on the PICOS framework for inclusion and exclusion are provided in Appendix B. The following criteria was used for initial screening of the studies.

[1] UN Digital Library, 3ie, United Nations Development Program, Food and Agriculture Organization, International Fund for Agricultural Development, World Bank (including World Bank Data Catalog, World Bank Policy Research, and Independent Evaluation Group), Agriculture and Food Organization, United States Agency of International Development, Consultative Group on International Agricultural Research, Economic Research Services of the United States Department of Agriculture, Agricultural Research Service of the United States Department of Agriculture, Inter-American Development Bank, African Development Bank Group (including Independent Development Evaluation of African Development Bank), AgEcon search, EconPapers of Research Papers in Economics.

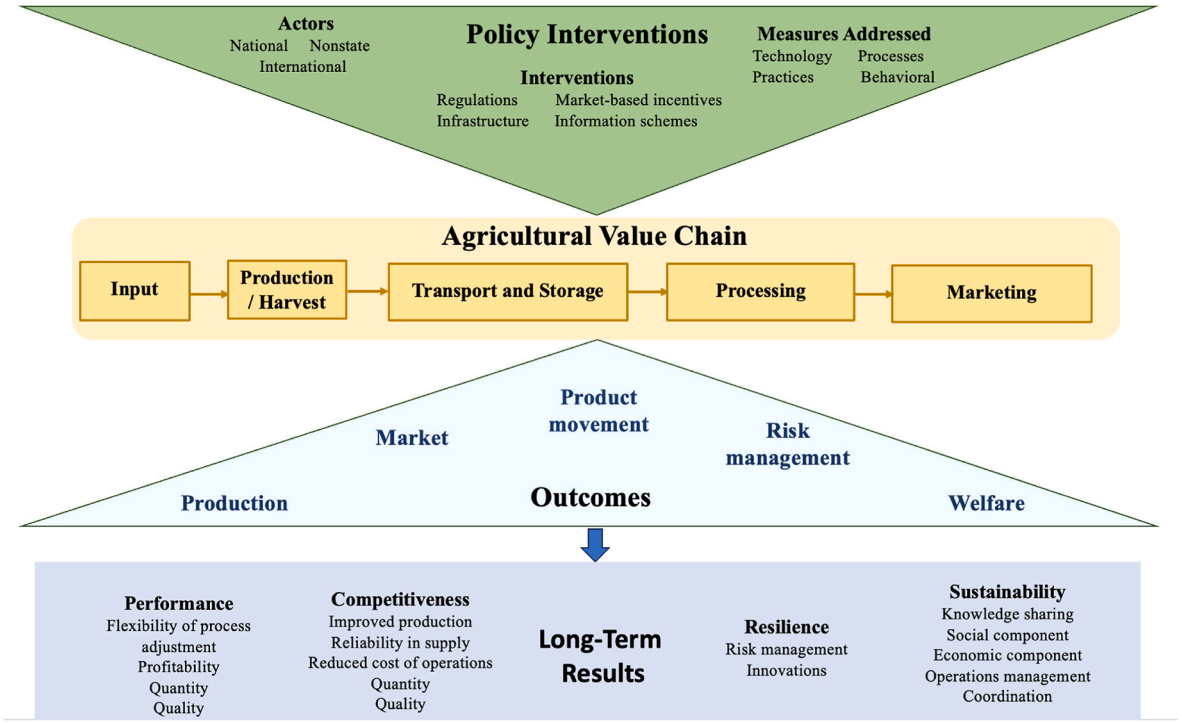


Fig. 1. Conceptual Model (Adapted from: [7,15,18], and [37]).

1) Studies had to be focused on the AVC, published in English between 2000 and 2022, and conducted in developing countries, as classified by World Bank [28].
2) A broad range of interventions, including national, regional, or local policies and programs occurring at any stage of the AVC are included (Table 1).
3) Studies had to include a comparison(s) of the impact(s) of the intervention; either a before and after comparison or compare control versus treatment groups.
4) Studies had to report the effect of AVC interventions in one or more of outcome categories (Table 2).
5) Studies had to use either quantitative or mixed (quantitative and qualitative) research methods.

3.3. Study exclusion criteria

Criteria used to exclude a study at the initial screening are as follows.

1) Studies were excluded if they related to urban agriculture, wildlife use, forest extraction, and wild fisheries.
2) Studies were excluded if they were based on simulation or hypothetical experiments and if they used non-robust methods, such as narrative analysis, policy summaries, or overviews with no detailed discussion of a comparison.
3) Interventions on small scale projects were also excluded, as they may not reflect long-term course of actions.

3.4. Study screening and data extraction

Search results were uploaded to Covidence, a web-based tool for managing and streamlining the systematic review process [29]. Study selection process involved: (1) title/abstract screening, (2) full-text evaluation of selected records, and (3) data extraction. Two reviewers independently screened each study for relevance and eligibility with a third reviewer resolving any discrepancies. General information, study design, context, and related evidence were extracted by two reviewers. A third reviewer checked for completeness and accuracy. This procedure ensured that study selection and data extraction followed the inclusion/exclusion criteria. General information included authors, countries covered, publication date, and publication type. Study design was the methodology used to collect and analyze data. Context related information included intervention description and specific outcomes reported. Intervention and outcomes were categorized and grouped based on main emerging themes to be included as EGM rows and columns.

3.5. Quality assessment of systematic reviews

The quality of the systematic reviews was assessed using a checklist of requirements (Appendix C) that included search and screening procedures, embedded biases, and methods used for combining study results [30]. Two reviewers independently evaluated each review using 12 questions from the checklist. Discrepancies were resolved by the two reviewers jointly. Points were then assigned to each review using a three-point scale, where a score of “three” indicated the requirements are fully satisfied, a “two” indicated requirements are partially satisfied, and a “one” indicated requirements are not satisfied. Four of these questions were evaluating study selection and inclusion bias, including independent screening by at least two reviewers and no restrictions on inclusion based on publication status. Two questions were addressing risk of bias while evaluating quality of included studies and their external validity. An additional two questions assessed reporting/-reviewer bias, including summary of the results, characteristics of the participants, interventions and outcomes, as well as the list of included studies. Finally, four questions were evaluating unit of analysis error or heterogeneity between included studies while weighting and combining the evidence. Scores were summed to classify the reviews into high, medium, or low-quality groups. The possible score for each study was between 12 and 36. Systematic reviews with scores from 14 to 21 were classified into low quality group, from 22 to 28 into medium quality group, and from 29 to 36 into high quality group.

4. Results

Details of the study selection process are shown in the Preferred

Table 1
Categories of intervention by stage of the agricultural value chain.

Intervention	Examples of Activities
Input Supply	
Input distribution network and management	Improved seed, fertilizers, herbicide and chemicals, provision of equipment
Information sharing and knowledge management	Technology transfer, crop selection and diversification, extension programs and activities, soil health management, weather information, knowledge translation
Institutional credit and subsidies	Sustainable credit availability and crop insurance through institutions, transactional subsidies
Land tenure and land improvement	Land rights, land capital investment, and efficient use of land resource
Water management	Irrigation practices, including groundwater usage. Water conservation activities and adoption of water saving technologies.
Production, Harvest, and Post-harvest	
Crop management services	Expert advice on pest and disease management, fertilizer application, irrigation management, harvest, storage, and distribution network for perishable and non-perishable produce
Dairy, meat, and livestock related services	Upgrading dairy plants and improving animal husbandry practices, policies on safe handling of animal origin waste
Transport and Storage	
Transport capacity (roads, refrigerated vehicles) and modernized storage	Rail, road, and sea transportation and storage services in temperature-controlled environment
Processing	
Value addition services	Grading, sorting, packaging, certification, crop processing, dairy and meat processing
Product traceability and food safety services	Adoption of food safety standards and good agricultural practices.
Marketing	
Contract farming and industry tie-up	Contract farming, activities aimed at vertical collaboration
Certification along value chain	Certification of products, including organic certification or fair trade.
Commodity markets and trading platforms	Trading platforms, commodity markets, local storage facility at marketplace, market access
Market information networks and channels	Price support (government and private companies), market access/farmers market, export markets
Governance and Inclusion	
Interventions around gender, poverty, and social issues	Programs aimed at reducing the poverty gap between urban and rural areas or between different socio-demographic groups of society.
Value chain coordination	Intervention to improve the governance and making sustainable value chain
Trade policies	Tariffs and trade agreements

Table 2
Categories and subcategories of intervention for various outcomes of the agricultural value chain.

Outcome	Outcome subcategories
Production	Yield
	Technology adoption
	Area under cultivation
	Technical efficiency
	Crop diversification
	Product quality
Market	Market access
	Agricultural profits
	Awareness/knowledge on market demand and prices
	Percent of crop sold
	Cross-border trade
	Inputs availability
Product movement in space and time	Produce procurement services
	Infrastructure (roads and facilities)
	Food waste management
	Food safety management
Risk management	Yield risk
	Price or market risk
	Contract risk
	Climate change risk
Welfare	Family income
	Food security and food access
	Nutritional security
	Poverty reduction
	Education and healthcare
	Employment conditions and opportunities
	Environmental conservation or pollution reduction
	Savings and consumption

Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart [31] in Fig. 2. After removing 322 duplicates, 4563 studies were screened based on titles and abstracts. At this initial screening state, common reasons for exclusion of studies were absence of robust study design, absence of comparison, and the study language being other than English. Of these 4563 studies 1041 were selected for full text screening. A total of 165 studies (Appendix D), of which 27 are systematic reviews and 138 are primary research studies, passed the full screening and are included in the EGM.

4.1. Overview of included studies

Only a few studies published in the 2000s were included (Fig. 3a). A noticeable increase in the number of included studies occurs post 2012. No studies published from 2000 through 2003 were selected for inclusion. The small number of studies published in 2022 is attributed to the search being conducted in April 2022. About 73 % of the studies were published as journal articles. The next largest category was reports (14 %), while the remaining 13 % were split between dissertations/theses, book chapters, working papers, and conference presentations.

Primary research. Geographic distribution of primary research by country is given in Fig. 3b. Some studies included interventions in only one country while others included multiple countries giving rise to more countries included than the number of primary research studies. The largest number of primary research studies were conducted in Africa and Asia and the smallest number of studies in the Americas. Primary research was conducted in all but six African countries. Ghana (n = 19) is the country with the largest number, closely followed by China (n = 17), Ethiopia (n = 15), and India (n = 9). Only 10 primary research studies were conducted in the Americas.

Approximately 81 % of the primary research used quasi-experimental methods in their study design, while 19 % used experimental methods. Statistical models used can be broadly grouped into two categories: those that compare means to identify treatment effects (14 %), and those that use more sophisticated approaches, such as instrumental variables (16 %), double hurdle models (13 %), panel data analysis (12 %), logit or probit models (12 %), ordinary least squares (10 %), and data envelopment analysis (9 %).

Interventions were primarily directed towards smallholder farmers. In over 63 % of the cases primary research interventions were initiated by national and local governments or driven by aid from other countries. The substantial government involvement can be attributed, in part, to a considerable proportion (70 %) of the interventions being a collaboration between international development organizations and local or national governments.

Systematic reviews. The 27 systematic reviews examined the impact of interventions on various AVC stages. Geographical distribution of the systematic reviews was not mapped since each review covers a wide geographic area. More than half of systematic reviews were published on government or donor organizations websites. Reviews discuss input supply (61 %), marketing (27 %), production, harvest, and post-harvest (5 %), governance and inclusion (5 %), processing (1 %), and transport and storage (1 %) stages of the AVCs (generally, each

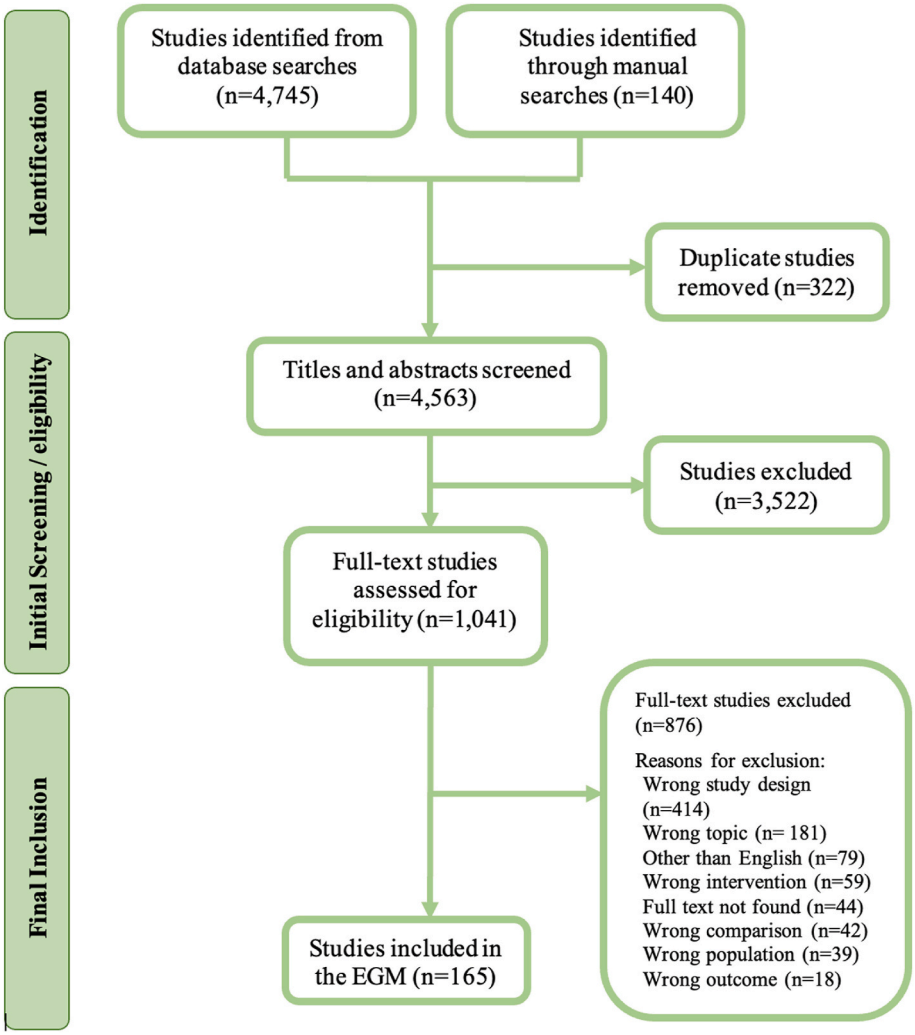


Fig. 2. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart presenting number of studies identified, reasons for exclusion and final number of studies included.

review discusses more than one intervention and/or stage). Among these reviews, 17 used descriptive methods and 10 used either meta- or meta-regression analysis to summarize the studies. Most of the systematic reviews did not report the quality assessment of included studies. Language bias (a common bias for reviews) was observed, as only seven systematic reviews incorporated papers published in multiple languages. Furthermore, nearly half of the reviews did not mention whether the screening was done by at least two reviewers. As a result, during the quality assessment, these reviews were labeled as having a reviewer bias. Seven reviews did not consider similarity (different scales, different settings, etc.) of the included studies when combining results, thus producing potential heterogeneity bias. All these factors were considered in systematic reviews' quality assessment stage, resulting in the classification of seven reviews being high, nine medium, and eleven low quality.

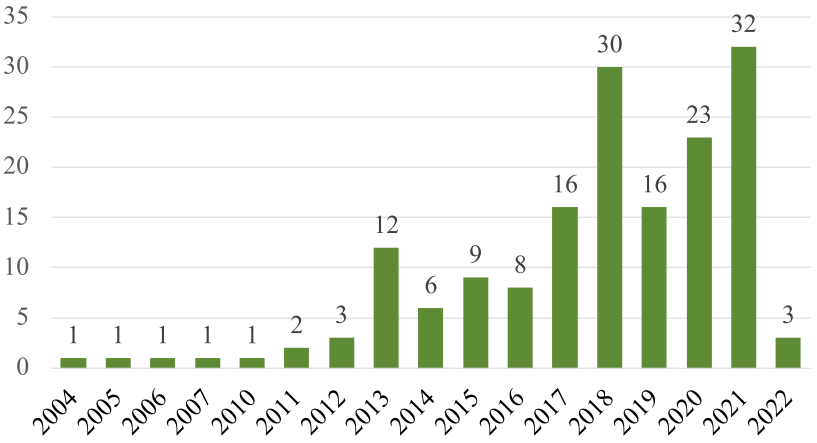
4.2. Gaps in AVC policy: interventions/outcomes

As previously noted, the EGM is a matrix of interventions (rows) and outcomes (columns) combinations with larger bubble sizes indicating a higher frequency of that combination. A particular study may be included in more than one cell. The color of the bubble indicates primary research and the quality of systematic reviews (see Fig. 4).

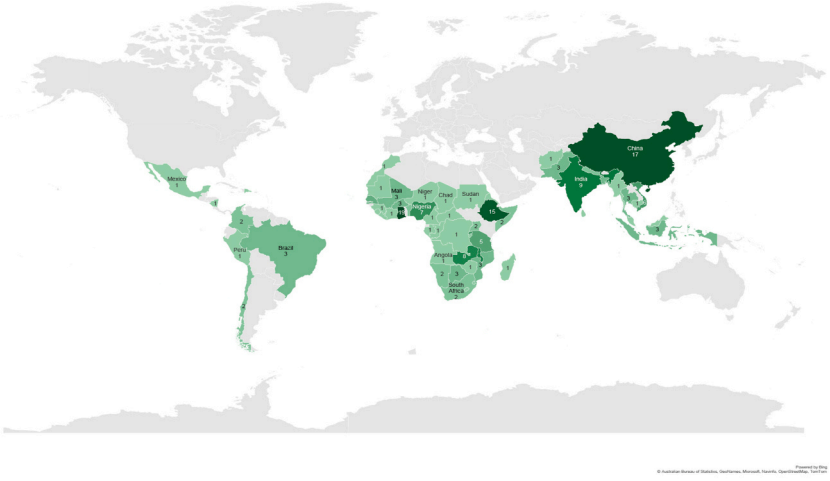
Interventions were mainly concentrated at the input supply stage of the AVC (Fig. 4), with the largest number of studies exploring institutional credit and subsidies, followed by input distribution networks, and knowledge sharing. Crop management in the production, harvest, and post-harvest stage and contract farming in the marketing stages follow the above three input stage interventions in terms of number of studies. The least studied interventions were in governance and inclusion, processing and transport, and storage stages of the AVCs. Two stages, processing and transport and storage, were each represented by only one systematic review, further indicating the limited number of primary research studies addressing these stages.

Interventions primarily focus on outcomes related to production, market, and welfare. Most populated intervention/outcome cells were found at the intersection of inputs and production outcomes. Besides the subcategories in the production outcome, agricultural profits (market outcome) and family income (welfare outcome) were highly populated part of the EGM. Empty cells, such as those at the intersection of transportation and storage, processing or some governance and inclusion categories with risk management outcomes, indicate gaps not addressed by previous studies. The only sub-category that was never mentioned as an outcome was food waste management.

There were no high-quality systematic reviews discussing interventions in governance and inclusion, processing, and production, harvest, and post-harvest stages of the AVCs. Medium quality systematic reviews were found for all stages of the AVC, except transport and storage which is represented by only one systematic review. Lack of systematic reviews for transport and storage may be a result of only one



a. Distribution of Included Primary Research and Systematic Reviews by Year.



b. Geographic Scope of Primary Research

Fig. 3. Included studies by year of publication and country.

intervention listed for this stage. The largest number (n = 9) of low-quality systematic reviews were focused on institutional credit and subsidies within the input supply stage. In terms of outcomes, emphasis of systematic reviews was on outcomes related to production, welfare, and market which is not surprising given the previous discussion on the number of primary research studies reporting these outcomes. Among outcome categories product movement in time and space was not discussed in any high-quality systematic reviews.

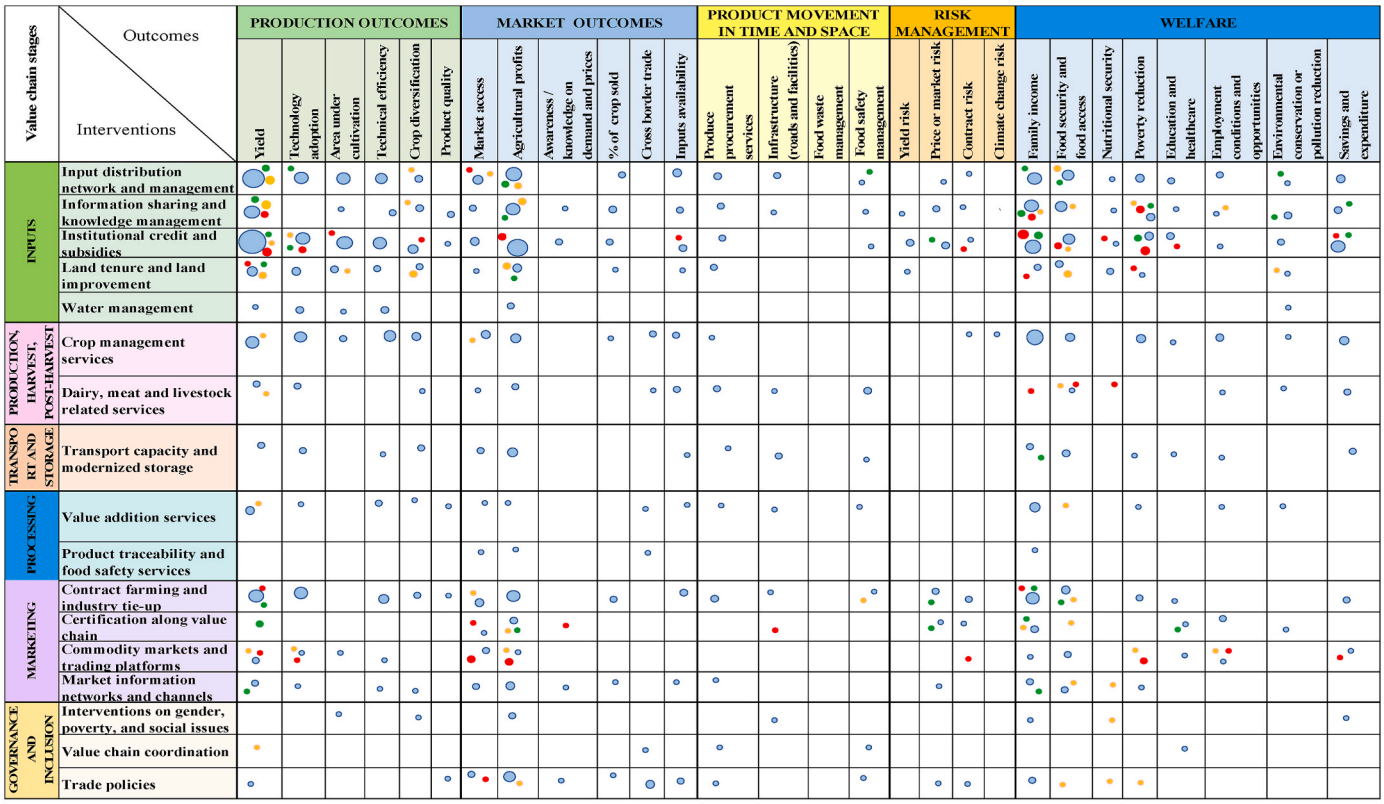
4.3. Results by interventions (rows)

The frequencies of interventions are presented as a heatmap by row which gives the number of outcomes for each specific intervention subcategory (Fig. 5). Shading in the cells is as follows. Green cells indicate the largest number of studies exploring the intervention subcategory. As the shade transitions to light green, yellow, and light red the number of studies in that intervention subcategory decreases. A red shade indicates no studies in that intervention and outcome intersection. The marginal increase in knowledge of another project which focuses on a green cell may not be as large as a project which focuses on a cell shaded red.

Among the various outcome subcategories (columns), yield is most likely to be mentioned as an outcome by the interventions in input, production, and marketing stages. The next most frequent subcategories of outcomes were agricultural profits followed by family income, food security, and market access. Interventions to promote commodity markets and trading platforms and market information networks and channels were more frequently examined as having outcomes related to market access and increased profits. Since the main group of stakeholders in this study are smallholder farmers, agricultural profits and family income may often be hard to separate or differentiate from each other. If combined, related outcomes in these two categories outweigh other individual categories along all rows except for land tenure, water management and value chain coordination.

4.4. Results by outcome categories (columns)

A heatmap by column which gives the numbers of interventions for each specific outcome subcategory is presented in Fig. 6. By examining this figure, one can observe what intervention may be more likely associated with a specific outcome. Institutional credit and subsidies was the intervention category impacting the largest number of outcomes, appearing in green in 16 out of 28 columns. In contrast, interventions in water management, value addition services, product traceability



Note: This table displays a matrix of interventions and outcomes combinations, indicating which interventions have resulted in which specific outcomes. The size of bubble increases based on the number of studies exploring the specific intervention / outcome combination. The same study may be included in more than one cell. Blue bubbles represent primary research studies; green bubbles represent high-quality systematic reviews, yellow represent medium quality, and red represent low quality.

Fig. 4. Evidence Gap Map of Policies Enhancing Agricultural Value Chain.
Note: This table displays a matrix of interventions and outcomes combinations, indicating which interventions have resulted in which specific outcomes. The size of bubble increases based on the number of studies exploring the specific intervention/outcome combination. The same study may be included in more than one cell. Blue bubbles represent primary research studies; green bubbles represent high-quality systematic reviews, yellow represent medium quality, and red represent low quality. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

services, certification along the value chain, market information networks and channels, and interventions related to gender, poverty, and social issues were not among top categories impacting any of the outcomes. Intuitively some of the interventions implemented in the marketing stage should be associated with outcomes in the market category, however studies did not report many market-related outcomes as indicated by red shading dominating cells in the intersection of marketing interventions market outcomes in Fig. 6.

5. Discussion

Agricultural value chains (AVCs) consist of multiple interrelated stages. An understanding of what interventions have been studied and their resulting outcomes in each stage of the AVC supports informed decision making about future areas of policy intervention, but also highlights underexplored research areas. The specific objective is to identify existing evidence on interventions in AVCs and to emphasize gaps in our knowledge that are identified by having none to only a few studies as indicated by an Evidence Gap Map (EGM). EGMs, which systematically collect and analyze studies, visually present information on the number of studies on a specific topic; therefore, allowing organizations and agencies to target knowledge gaps potentially leading to increased knowledge concerning policy interventions.

By stage the largest number of studies discussed interventions in the input, production, harvest and post-harvest, and marketing stages. Whereas the stages with the smallest number of studies were transport and storage, processing, and governance and inclusion. By outcome subcategories the largest number of studies were in the yield, agricultural profits, family income, and technology adoption. Outcomes related to risk management and product movement in time and space had the smallest frequency of occurrence, providing opportunities for future research. Climate risk was mentioned by only one study and food waste was the only outcome subcategory that was not mentioned by any of the studies.

Examples of where knowledge gaps exist and the need for additional studies include the following. Approximately 30 % of food may be wasted in developing countries [32] where food supply may be limited giving rise to malnutrition and food insecurity. The gap in studies addressing food waste indicates inattention to this important topic. Interventions in gender, poverty, and social issues would improve our knowledge base translating into impactful results since in developing countries poverty rates are high and there is a large female involvement in agricultural activities [33]. There is a gap in the knowledge base on interventions related to product traceability, which was mentioned by only two studies. Given the global food chain and health issues [34,35] additional research on traceability may be warranted.

Interventions focused on input stage were linked with outcomes related to market access more than marketing stage interventions. One possible explanation is that smallholder farmers, which is typical in developing countries, usually have plot sizes between 1.5 and 3 ha [36].

Value chain stages	Interventions \ Outcomes	PRODUCTION OUTCOMES						MARKET OUTCOMES						PRODUCT MOVEMENT IN TIME AND SPACE				RISK MANAGEMENT				WELFARE									
		Yield	Technology adoption	Area under cultivation	Technical efficiency	Crop diversification	Product quality	Market access	Agricultural profits	Awareness / knowledge on market demand	% of crop sold	Cross border trade	Inputs availability	Produce procurement	Infrastructure (roads and facilities)	Food waste management	Food safety management	Yield risk	Price or market risk	Contract risk	Climate change risk	Family income	Food security and food access	Nutritional security	Poverty reduction	Education and healthcare	Employment conditions and opportunities	Environmental conservation or pollution reduction	Savings and expenditure
INPUTS	Input distribution network and management	31	12	8	6	4	0	7	21	0	2	0	4	3	2	0	2	0	1	1	0	13	11	1	4	2	2	2	4
	Information sharing and knowledge management	23	14	1	3	4	2	2	14	1	3	0	2	2	1	0	1	1	2	1	0	16	8	1	9	1	2	4	4
	Institutional credit and subsidies	41	15	17	8	6	1	4	26	2	2	0	4	2	0	0	1	3	5	2	0	24	11	2	10	4	1	2	9
	Land tenure and land improvement	11	4	4	3	5	0	1	4	0	1	0	1	2	0	0	0	1	0	0	0	3	6	2	2	0	0	2	0
	Water management	1	2	1	3	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
PRODUCTION, HARVEST, POST-HARVEST	Crop management services	8	5	1	5	2	0	2	4	0	1	1	1	1	0	0	0	0	0	1	1	9	3	0	3	1	2	1	3
	Dairy, meat and livestock related services	3	2	0	0	1	0	1	3	0	0	1	2	2	1	0	3	0	0	0	0	1	3	1	0	0	1	1	2
TRANSPORT AND STORAGE	Transport capacity, modernized storage, and training for good storage	2	2	0	1	2	0	2	5	0	0	0	1	1	2	0	1	0	0	0	0	3	3	0	1	1	1	0	2
POCESSING	Value addition services	5	1	0	2	1	1	1	1	0	0	1	1	1	1	0	1	0	0	0	0	6	1	0	1	0	1	1	0
	Product traceability and food safety services	0	0	0	0	0	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
MARKETING	Contract farming and industry tie- up	16	8	0	5	2	1	5	9	0	2	0	3	3	1	0	2	0	3	3	0	9	6	0	2	1	0	0	2
	Certification along value chain	2	0	0	0	0	0	2	5	1	0	0	0	0	1	0	0	0	2	1	0	5	1	0	0	2	2	1	0
	Commodity markets and trading platforms	4	3	1	1	0	0	5	4	0	0	0	0	0	0	0	0	0	0	1	0	1	2	0	3	1	3	0	2
	Market information networks and channels	3	1	0	1	1	0	2	4	1	1	0	1	1	0	0	0	0	1	0	0	3	3	1	1	0	0	0	0
GOVERNANCE AND INCLUSION	Interventions on gender, poverty, and social issues	0	0	1	0	1	0	0	2	0	0	0	0	0	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1
	Value chain coordination	1	0	0	0	0	0	0	0	0	0	1	0	1	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0
	Trade policies	1	0	0	0	0	1	3	8	1	1	4	2	1	0	0	1	0	1	1	0	2	1	1	1	0	0	0	0

Fig. 5. Number of Studies Heatmap by Rows.
Note: Heatmap sorted by row highlights the frequency of occurrence by reporting the number of outcomes for each intervention category, where dark green represents the largest frequency along the row and the red represents the lowest. For example, the second intervention category "Information sharing and knowledge management" resulted in change in the "Yield" outcome category in 23 studies. Being the highest number in that row, means that for this intervention yields occurred more than any other outcome. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Due to their limited output, these farmers may not be able to enter the market as individual sellers or may not be willing to bear the costs of certification. However, they may collaborate with processors to supply the required quality of produce. This may be supported by the findings among interventions related to the marketing stage of the value chain; contract-farming sub-category has more studies than the other three sub-categories. These finding highlight the nature of smallholder farming and their dependency on processors or retailers.

6. Conclusions

This study uses an EGM to identify stages of AVCs that are overlooked in the literature concerning policy interventions and their outcomes. Transportation and storage, for example, are major infrastructural challenges in developing countries, as there is a lack of studies addressing policies which may improve this stage of the food value chain. As such, the literature can only provide limited evidence concerning how such policies may improve AVCs. The EGM identifies other such gaps in policy interventions at each stage of AVC. If policymakers are constrained to certain types of interventions, it is helpful to understand what are the most likely outcomes that were meant to be achieved through those interventions. The evidence shows that institutional credit and subsidies may be influential interventions because they were paired with more outcome categories than any other intervention. If the policy goal is to impact yield, more likely interventions should be those related to the input supply stage, since studies show that nearly all sub-categories in input supply stage interventions had the highest frequency of being associated with yield.

It is stressed this is not a complete listing but just some examples of how the EGM can be used to guide funding and research. While analyzing EGM results, it is important to note that the degree of attention and emphasis on various interventions in the studies could be dictated by initiators of those interventions. Government and international organizations tend to initiate and support interventions in the input supply stage, such as subsidies, seed provision, and extension services. Because donor organizations and governments require publicity for the implemented projects, they would encourage publications regarding these interventions and resulting outcomes. On the other hand, interventions, such as certification would be typically handled by farmers and processors, who may not publish their outcomes. Consequently, there could be a larger number of publications examining the input stage compared to the other stages of the value chain.

7. Study limitations

Although EGM allows better understanding of gaps related to interventions implemented along AVCs, the results of this study are time and language bound. Information was drawn only from literature published in English and after 2000. In addition, one should keep in mind that any review reflects literature made available to the community. Many studies may not be documented leading to their exclusion. The degree of attention and emphasis on the various stages of the value chain differs depending on the specific context and challenges confronted by different agricultural systems and supply chains. EGMs do not attempt to determine the effect a policy intervention had on an outcome. As such, readers are cautioned to use the EGM as frequency of studies and not the effect of a policy intervention. Finally, included studies had to meet certain requirements, obviously changing the requirements may change the results.

Value chain stages	Interventions \ Outcomes	PRODUCTION OUTCOMES						MARKET OUTCOMES						PRODUCT MOVEMENT IN TIME AND SPACE					RISK MANAGEMENT				WELFARE								
		Yield	Technology adoption	Area under cultivation	Technical efficiency	Crop diversification	Product quality	Market access	Agricultural profits	Awareness / knowledge on market demand	% of crop sold	Cross border trade	Inputs availability	Produce procurement	Infrastructure (roads and facilities)	Food waste management	Food safety management	Yield risk	Price or market risk	Contract risk	Climate change risk	Family income	Food security and food access	Nutritional security	Poverty reduction	Education and healthcare	Employment conditions and opportunities	Environmental conservation or pollution reduction	Savings and expenditure
INPUTS	Input distribution network and management	31	12	8	6	4	0	7	21	0	2	0	4	3	2	0	2	0	1	1	0	13	11	1	4	2	2	2	4
	Information sharing and knowledge management	23	14	1	3	4	2	2	14	1	3	0	2	2	1	0	1	1	2	1	0	16	8	1	9	1	2	4	4
	Institutional credit and subsidies	41	15	17	8	6	1	4	26	2	2	0	4	2	0	0	1	3	5	2	0	24	11	2	10	4	1	2	9
	Land tenure and land improvement	11	4	4	3	5	0	1	4	0	1	0	1	2	0	0	0	1	0	0	0	3	6	2	2	0	0	2	0
	Water management	1	2	1	3	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
PRODUCTION, HARVEST, POST-HARVEST	Crop management services	8	5	1	5	2	0	2	4	0	1	1	1	1	0	0	0	0	0	1	1	9	3	0	3	1	2	1	3
	Dairy, meat and livestock related services	3	2	0	0	1	0	1	3	0	0	1	2	2	1	0	3	0	0	0	0	1	3	1	0	0	1	1	2
TRANSPORT AND STORAGE	Transport capacity, modernized storage, and training for good storage	2	2	0	1	2	0	2	5	0	0	0	1	1	2	0	1	0	0	0	0	3	3	0	1	1	1	0	2
POCESSING	Value addition services	5	1	0	2	1	1	1	1	0	0	1	1	1	1	0	1	0	0	0	0	6	1	0	1	0	1	1	0
	Product traceability and food safety services	0	0	0	0	0	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
MARKETING	Contract farming and industry tie- up	16	8	0	5	2	1	5	9	0	2	0	3	3	1	0	2	0	3	3	0	9	6	0	2	1	0	0	2
	Certification along value chain	2	0	0	0	0	0	2	5	1	0	0	0	0	1	0	0	0	2	1	0	5	1	0	0	2	2	1	0
	Commodity markets and trading platforms	4	3	1	1	0	0	5	4	0	0	0	0	0	0	0	0	0	0	1	0	1	2	0	3	1	3	0	2
	Market information networks and channels	3	1	0	1	1	0	2	4	1	1	0	1	1	0	0	0	0	1	0	0	3	3	1	1	0	0	0	0
GOVERNANCE AND INCLUSION	Interventions on gender, poverty, and social issues	0	0	1	0	1	0	0	2	0	0	0	0	0	1	0	0	0	0	0	0	1	0	1	0	0	0	0	1
	Value chain coordination	1	0	0	0	0	0	0	0	0	0	1	0	1	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0
	Trade policies	1	0	0	0	0	1	3	8	1	1	4	2	1	0	0	1	0	1	1	0	2	1	1	1	0	0	0	0

Fig. 6. Number of Studies Heatmap by Columns.
Note: Heatmap sorted by column shows what category of intervention corresponds to the highest number of studies for each particular outcome where dark green represents the largest frequency along the column and the red represents the lowest. For example, second intervention category called "Information sharing and knowledge management" resulted in change in outcome called "Yield" in 23 studies. This is the third largest number along that column. The highest number is 41, which means "Yield" occurred more frequently under intervention called "Institutional credit and subsidies". (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

CRediT authorship contribution statement

Anna Yeritsyan: Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Conceptualization. **Elizabeth Tabares:** Writing – review & editing, Supervision, Methodology, Data curation, Conceptualization. **Ariun Ishdorj:** Writing – review & editing, Validation, Supervision, Funding acquisition, Data curation. **James Mjelde:** Writing – review & editing, Validation, Formal analysis, Data curation. **Vikas Mishra:** Methodology, Conceptualization. **Margaret Foster:** Database searches. **Roger Norton:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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